

Post-Streptococcal Glomerulonephritis (PSGN)



1. Nephritogenic M-type strep

WHAT YOU SEE

Sunglasses men, 'M-type nephritogenic'

WHAT IT ENCODES

PSGN follows infection with specific nephritogenic strains of group A beta-hemolytic Streptococcus (certain M protein types). Only nephritogenic strains trigger glomerulonephritis; rheumatic fever and PSGN are caused by different strains and do not coexist.

BOARD TRAP

Antibiotic treatment of strep prevents rheumatic fever but does NOT prevent PSGN — the immune-complex process is already set in motion.

2. 2 weeks ago strep throat / 6 weeks skin

WHAT YOU SEE

Boy with timeline 'pharyngitis 2 wk / skin 6 wk'

WHAT IT ENCODES

Classic latency: ~1–2 weeks after pharyngitis or ~3–6 weeks after impetigo/skin infection. This lag distinguishes PSGN from IgA nephropathy, which causes synpharyngitic hematuria (within 1–3 days of infection).

BOARD TRAP

Hematuria DURING the infection (synpharyngitic, days) is IgA nephropathy; PSGN appears 1–6 weeks AFTER the infection resolves.

3. Hematuria + RBC casts

WHAT YOU SEE

Urine cup, 'hematuria + RBC casts'

WHAT IT ENCODES

Nephritic syndrome: cola-colored urine, dysmorphic RBCs with RBC casts, mild-to-moderate proteinuria, hypertension, edema, and oliguric AKI. The active urine sediment confirms a glomerular bleeding source.

BOARD TRAP

RBC casts mean a glomerular (nephritic) source — isolated hematuria without casts or dysmorphic cells suggests a urologic, not glomerular, cause.

4. Periorbital / facial puffiness

WHAT YOU SEE

Boy with 'puffy bital puffiness'

WHAT IT ENCODES

Salt and water retention from reduced GFR cause edema, classically periorbital puffiness on waking, plus hypertension. This volume overload, not hypoalbuminemia, drives the edema in nephritic disease.

BOARD TRAP

Nephritic edema is from sodium/water retention and a low GFR — it is not nephrotic edema from urinary protein/albumin loss.

5. Subepithelial humps — EM

WHAT YOU SEE

Purple germ, 'subepithelial humps — EM'

WHAT IT ENCODES

Electron microscopy shows large dome-shaped subepithelial immune-complex deposits ('humps'). Light microscopy shows diffuse hypercellular endocapillary proliferation; IF shows granular 'starry sky' IgG and C3 along the GBM and mesangium.

BOARD TRAP

Subepithelial humps are PSGN; subepithelial spikes/dome deposits with PLA2R point to membranous nephropathy — different deposit morphology and clinical syndrome.

6. Hypercellular (endocapillary proliferation)

WHAT YOU SEE

Bouncy castle, 'hypercellular'

WHAT IT ENCODES

Light microscopy reveals diffuse proliferative, hypercellular glomeruli packed with neutrophils and proliferating endocapillary cells. This exudative pattern explains the active nephritic sediment and transient renal dysfunction.

BOARD TRAP

Hypercellularity here is endocapillary/exudative — not the crescentic (extracapillary) proliferation of RPGN, though severe PSGN can rarely form crescents.

7. Low complement (C3 ↓)

WHAT YOU SEE

'C3 ~25%' cultures gauge

WHAT IT ENCODES

Complement is consumed: serum C3 falls (low) and normalizes within ~6–8 weeks. Persistently low C3 beyond 8 weeks should prompt reconsideration (e.g., MPGN, C3 glomerulopathy, or lupus).

BOARD TRAP

PSGN's low C3 should normalize by 8 weeks — persistent hypocomplementemia points away from PSGN toward MPGN/C3 GN or lupus.

8. Streptozyme 80–95%+ / ASO, anti-DNase B

WHAT YOU SEE

Trophies 'streptozyme 80-95%'

WHAT IT ENCODES

Evidence of recent strep infection: elevated ASO titer (better after pharyngitis) and anti-DNase B (better after skin infection). The streptozyme panel detecting multiple antibodies has the highest sensitivity (80–95%).

BOARD TRAP

After skin infection (impetigo) ASO is often low — order anti-DNase B, which is the more sensitive marker for streptococcal pyoderma.

9. Treatment is supportive

WHAT YOU SEE

Trash can 'no role: with cyclophosph'

WHAT IT ENCODES

Management is supportive: sodium/water restriction, loop diuretics, and blood-pressure control while the self-limited disease resolves. Antibiotics treat any persistent infection but do not alter the renal course.

BOARD TRAP

Immunosuppression (steroids/cyclophosphamide) has no role in typical PSGN — reserve it only for rare biopsy-proven crescentic/RPGN progression.

10. Children resolve / elderly worse

WHAT YOU SEE

Elderly on stretcher, 'azotemia 60%, ESKD'

WHAT IT ENCODES

Prognosis is excellent in children (>95% full recovery). Adults, especially the elderly, fare worse: higher rates of azotemia, persistent hypertension/proteinuria, and progression to CKD/ESKD.

BOARD TRAP

Don't quote the benign pediatric prognosis for an elderly adult — older patients have substantial rates of azotemia and incomplete recovery.

11. Subclinical 4–5x more common

WHAT YOU SEE

Faces, 'subclinical 4-5x more common'

WHAT IT ENCODES

For every clinically overt case there are several subclinical cases detected only by microscopic hematuria and transient complement changes. This is why family screening after an index case can reveal additional affected individuals.

BOARD TRAP

Most PSGN is subclinical — absence of gross hematuria/edema does not exclude glomerular involvement after a documented strep infection.

12. ANCA-MPO 10% (overlap)

WHAT YOU SEE

'ANCA-MPO 10%' tag

WHAT IT ENCODES

A subset of post-streptococcal/post-infectious GN patients are ANCA-positive, which can predict a more aggressive, sometimes crescentic course. Recognizing overlap matters because it may justify a biopsy and immunosuppression.

BOARD TRAP

Persistent or rapidly worsening renal function in apparent PSGN warrants ANCA testing and biopsy — an overlap vasculitis changes management toward immunosuppression.
